

Nuclear Engineering

Nuclear engineers develop the processes, instruments and systems for national laboratories, private industry and universities that derive benefits from nuclear energy for society. As a nuclear engineer, you may solve challenges in the areas of consumer and industrial power, space exploration, world food and water supply, environmental protection, medicine and transportation.

In other words, this branch of engineering deals with the technology of harnessing energy contained in the atom generally by fission or fusion.

Key skills required

As a nuclear engineer, you can work in various divisions like electronics, reactor control, accelerator, analytical chemistry, applied physics, atomic fuels, design and manufacturing, chemical engineering, cryo technologies, waste management, food technology, fuel reprocessing, health, heavy water reactor, laser and plasma, amongst others. The nature of work includes R&D, hardware development, software reliability testing, system analysis and control, etc.

Recruiters look for a good understanding of the subject and evaluate candidates based on their in-depth knowledge of nuclear science and basic skills like mental alertness. They also consider internships and research projects candidates might have participated in.

Who is hiring?

The biggest recruiter is the Department of Atomic Energy, government of India, with a large number of organisations running under it. These include BARC, Trombay; IGCAR, Kalpakkam; and Nuclear Power Corporation of India Limited (NPCIL), Mumbai.

Besides, you can get placed at centres like Atomic Minerals Directorate for Exploration and Research (Hyderabad), Bharatiya Nabhiya Viduyt Nigam Ltd (Kalpakkam), Board of Radiation and Isotope Technology (Mumbai), Raja Ramanna Centre for Advanced Technology (Indore), Heavy Water Board (Mumbai), Nuclear Fuel Complex (Hyderabad), Uranium Corporation of India Ltd (Jaduguda) and Variable Energy Cyclotron Centre (Kolkata).

Power Electronics

Power electronics has a very high potential as now it has shifted from a mere IT product (UPS, inverter, etc) to powering the whole plant and also other sectors like lifts and colour photo machines.

Power electronics is very important in the industrial, energy and automotive sectors, among others. The industrial sector offers job opportunities in the motor drives area. In the energy sector, it offers opportunities in companies that work on solar and wind energy conversion and power system compensation. The automotive sector offers job opportunities in the design of various automobile sub-systems, and electric and hybrid vehicles. Many of these are related to energy efficiency and usage. In the present energy scenario, opportunities and prospects for power electronics engineers are very good.

Key skills required

The industry requires professionals with varied skillsets to cater to product design and development, project engineering, product management, marketing and sales. The companies in this sector also look for professionals having a superior understanding of electronics, instrumentation, electrical, mechanical and power engineering.

The power sector offers huge opportunities for engineering professionals with multidisciplinary interests such as magnetics and mechanics. Technically satisfying careers can also be pursued in the fields of technical marketing, design engineering, project management, and integration and servicing.

As an electronics professional, you may pick a role in the power circuit design section of a solutions providing firm or in the maintenance, instrumentation and control sections of any power plant.

Coming to the basic criteria for entry, diploma holders and engineering graduates may try their luck here. Power industry recruiters also look for postgraduates and doctorates for some fields requiring specialised knowledge, such as R&D.

Major recruiters

The power sector comprises a wide range of firms from fairly large companies to small and medium-scale businesses. Major players include NTPC, BHEL, Reliance Tata Power, Sterlite Industries (India), Jindal India Thermal Power and Lanco Infratech.